

Priyams Ratna Bajracharya

Data Engineer | Python & Automation

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PROFESSIONAL SUMMARY

Computer Engineering graduate specializing in data pipeline development, ETL design, dimensional modeling, and workflow orchestration with PostgreSQL and Apache Airflow. Certified via the Leapfrog Connect Data Engineering Program, with hands-on experience building production-grade data warehouses from raw ingestion to verified, idempotent loads and downstream Power BI dashboards.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML, CSS

Data Engineering: PostgreSQL, Schema Design & Normalization, Star Schema / Dimensional Modeling, ETL, Incremental & Idempotent Loading, Window Functions, Query Optimization, Airflow

Data & Analysis: pandas, NumPy

Visualization: Power BI (DAX, drill-through), Matplotlib, React Dashboards

Tools: Git, GitHub, Jupyter Notebook, VS Code, Django, DBeaver

AI-Assisted Development: Use of Claude for code review, debugging, and pair-programming; comfortable adopting AI coding assistants (e.g. Claude Code) as core dev tooling

PROJECTS

Healthcare Analytics Data Warehouse — *Completed*

- Built an analytics-ready warehouse from 1,163 patients / 61K+ source encounters using a two-database architecture (healthcare_oltp → healthcare_dw); star schema with 6 dimensions, 1 fact table, and 2 bridge tables resolving encounter-to-diagnosis and encounter-to-procedure many-to-many relationships
- Python ETL (extract → transform → quality gate → load) orchestrated via a staged Apache Airflow DAG — parallelized dimension loads, per-task retry on the fact table; verified live: full load (40K+ fact rows), incremental watermark-driven load, and idempotency re-runs producing zero duplicate inserts
- Pre-load data quality gates (row count, null-key, referential integrity, non-negative cost checks) and Power BI dashboard (2 pages) answering 4 locked business questions on cost and utilization

github.com/Priyams-Bajracharya/healthcare-analytics-data-warehouse

Ride-Hailing Analytics Data Warehouse — *Completed*

- Normalized a flat dataset into a 5-table OLTP schema, then a star schema (fact_trips + 6 dims); batch loaders validated via EXPLAIN ANALYZE/indexing on 1M+ rows

github.com/Priyams-Bajracharya/Data-Engineering-Course

Adaptive Traffic Signal Controller (ATSC)

- Co-developed a DQN (Deep Q-Network) agent to optimize traffic signal timing at a real Kathmandu intersection (Baneshwor), simulated and validated in SUMO

AQI Forecast — Kathmandu

- Built a full-stack Django app forecasting Kathmandu air quality using a Random Forest model, from raw sensor CSV ingestion through data cleaning, model tuning, and UI integration

WORK EXPERIENCE

Frontend Developer Intern — Outlines Research and Development Pvt. Ltd. *Nov 2025 – Feb 2026*

- Integrated REST APIs into the frontend as primary responsibility — built request/response handling and debugged data/integration bugs using browser DevTools (network calls, headers, payloads)
- Built interactive React dashboards on top of the integrated data, translating business requirements into visualizations
- Contributed to frontend architecture decisions; collaborated cross-functionally to hit milestones

EDUCATION

Bachelor of Engineering in Computer Engineering — Kathmandu Engineering College, TU *2022 – 2026*

Graduated with Distinction — Final Aggregate: 82.90%

Higher Secondary Education (+2 Science) — St. Xavier's College, Kathmandu *2021*

CERTIFICATIONS & TRAINING

- Leapfrog Connect – Data Engineering Program (Completed) — PostgreSQL, ETL, Dimensional Modeling, Airflow
- DataCamp – Associate Data Engineer in SQL (Certified) · DataCamp – Data Engineer in Python (Certified)